Žan Grad – Curriculum Vitae

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Basic information

- Nationality: Slovenian
- Place and date of birth: Celje (Slovenia), October 6, 1993
- Current institution: KU Leuven, Belgium

Education

Doctoral degree in Mathematics (Ph. D.) Graduated June 27, 2025

Instituto Superior Técnico, University of Lisbon

- Member of Centre for Mathematical Analysis, Geometry, and Dynamical Systems
- Topic: *Fundamentals of Lie categories and Yang–Mills theory for multiplicative Ehresmann connections*.
- Thesis awarded with honors and distinction (*summa cum laude*).
- Advisers: Ioan Mărcuț and Pedro Resende.

Master's degree in Mathematics (M. Sc.) Graduated September 10, 2020

University of Ljubljana, Faculty of Mathematics and Physics

- Topic: *Gauge field theory, Yang–Mills–Higgs equations and spin structures on Lorentzian manifolds*.
- Thesis awarded with the Prešeren award of University of Ljubljana.
- Adviser: Sašo Strle.

Bachelor's degree in Physics (B. Sc.) Graduated July 7, 2017

University of Ljubljana, Faculty of Mathematics and Physics

- Topic: *Wave function collapse upon measuring the number of photons in a cavity.*
- Adviser: Tomaž Rejec.

Employment

Postdoctoral researcher in Pure Mathematics Sep 2025 – Ongoing

KU Leuven, Belgium

- Field of research: Poisson geometry and Lie theory.
- Adviser: Marco Zambon.

Assistant professor of mathematics Sep 2020 – Sep 2021

University of Ljubljana, Faculty of Mathematics and Physics

- Employed full time for two semesters, with an average load of 10 hours of teaching per week.
- Held exercise sessions, constructed written exams, graded students.
- Taught two master-level courses: Analysis on manifolds and Differential geometry.
- Taught five bachelor-level courses: Analysis IV for financial mathematicians, Mathematics 2 for physicists, Mathematics 4 for physicists, Basics of mathematical analysis for computer engineers, Mathematics for pharmacists.

Recorded exercise sessions for these courses are available on my personal webpage, in the teaching section.

Publications

[1] Žan Grad, *Fundamentals of Lie categories*, J. Noncommut. Geom. **19** (2025), no. 1, 211—248, DOI 10.4171/JNCG/563. (arXiv).

[2] ———, *Covariant derivatives in the representation-valued Bott–Shulman–Stasheff and Weil complex.* (arXiv).

[3] ———, *Yang–Mills theory for multiplicative Ehresmann connections.* In preparation.

Talks

- Planned: *Yang–Mills theory for multiplicative Ehresmann connections*, Higher Geometric Structures along the Lower Rhine XIX, University of Cologne, September 25-26 2025.
- *Yang–Mills theory for multiplicative Ehresmann connections*, University of Illinois Urbana–Champaign (Symplectic and Poisson Geometry Seminar), April 28, 2025.
- *Double complexes of representation-valued forms*, Universität zu Köln, Poisson Geometry Seminar, February 5, 2025.
- *Double complexes of representation-valued forms*, JMU Würzburg, Oberseminar Deformationsquantisierung und Geometrie, January 31, 2025.
- *The exterior covariant derivative of multiplicative forms*, IMPA, Rio de Janeiro, November 14, 2024.
- *Crash course on Symplectic fibrations*, Radboud University Nijmegen, April 2024.
- *Mini course on Lie categories*, Lie groupoid and algebroid week in Coimbra II, July 2023.
- *Lie categories*, Geometry Seminar of Lisbon, Instituto Superior Técnico, Universidade de Lisboa, May 9, 2023.
- *Lie categories*, PhD seminar of Radboud University Nijmegen, April 2023.
- Two talks on *Lie categories*, Topology Seminar of Faculty of Mathematics and Physics, University of Ljubljana, March 20 and March 27, 2023.
- A series of lectures on Lie groupoids, Instituto Superior Técnico, Universidade de Lisboa, Spring semester 2021/22.
- *Mini course on Minkowski space and the theory of relativity*, Summer school of Mathematics and Physics at University of Ljubljana, August 2019.

In the spring semester of 2024, I organized an online seminar on the integration of Lie algebroids to Lie groupoids, where we discussed and proved the well-known result by Crainic and Fernandes stating the precise obstructions to the integrability of Lie algebroids. This seminar was co-organized by fellow PhD students from University of Coimbra and Radboud University.

Conferences and Workshops

The following list contains attended and planned conferences and workshops.

- Higher Geometric Structures along the Lower Rhine XIX, September 25-26 2025 (Cologne).
- Interactions of Poisson Geometry, Lie Theory and Symmetry, June 30-July 4, 2025 (Lisbon).
- Higher Geometric Structures along the Lower Rhine XVIII, January 23-24, 2025 (Nijmegen).
- XI Workshop on Poisson Geometry and Related Topics, November 4-8, 2024 (Manaus).
- XXXII International Fall Workshop on Geometry and Physics, September 2-5, 2024 (Coimbra).
- Poisson School & Conference 2024, July 1-12, 2024 (Naples).
- Lie groupoid and algebroid week III, June 3-7, 2024 (Coimbra).
- Higher Geometric Structures along the Lower Rhine XVII, May 2-3, 2024 (Bonn).
- Lie groupoid and algebroid week II, July 10-14, 2023 (Coimbra).
- The many interactions of symplectic and Poisson geometry, June 19-23, 2023 (Paris).
- Lie groupoid and algebroid week I, September 26-30, 2022 (Coimbra).
- Poisson School & Conference 2022, July 18-29, 2022 (attended online).

Posters

- The exterior covariant derivative of multiplicative forms for XI Workshop on Poisson Geometry and Related Topics (Poisson in the Amazon), November 2024.

Grants

- COST CaLISTA European grant for short-term scientific missions. This grant enabled me to visit Ioan Mărcuț in person in Nijmegen, in the period of April and May 2024.
- PhD grant UI/BD/152069/2021 by Fundação para a Ciência e a Tecnologia (FCT), Portugal.

Awards

- *Prešeren award of Faculty of Mathematics and Physics, University of Ljubljana*, for master's thesis *Gauge field theory, Yang–Mills–Higgs equations and spin structures on Lorentzian manifolds*, 2020.

Mentoring

- Master's student: *Matthijs Lau*, Radboud University Nijmegen. Topic: *Symplectic groupoid fibrations.* Graduated in June 2025. This is a joint supervision with Ioan Mărcuț.

Other

Physics

I am interested in applying modern mathematical methods to describe the theory of general relativity, quantum mechanics, many-body quantum mechanics, quantum field theory, classical mechanics, classical field theory (hydrodynamics, electromagnetism, elastomechanics and gauge field theory).

Languages

Native speaker of Slovenian language. I am fluent in both written and spoken English (C2 proficiency).

Programming languages

I possess advanced knowledge of languages of the web: PHP, Javascript, HTML, CSS. Principles of backend and frontend frameworks, together with concrete advanced implementations of frameworks Laravel (PHP) in VueJS (Javscript). To support myself, I worked a part-time job as a full-stack web developer between 2010 and 2016. I also have basic knowledge of Mathematica and Python and some familiarity with C.